

8. The most common understanding of the term 'heavy metal' is a metallic element which is toxic and has a high density, atomic number and relative atomic mass. The definitions used vary depending on the context. In metallurgy, for example, a heavy metal is defined on the basis of density, whereas in physics the distinguishing factor is atomic number. A chemist would likely be more concerned with chemical behaviour. More specific definitions have been published but none of these have been widely accepted.

Despite this lack of agreement, the term is widely used in science. Heavy metals are sometimes defined as metals with a density greater than 5 g/cm^3 . They are often highly toxic or damaging to the environment. Chromium, arsenic, cadmium, mercury and lead have the greatest potential to cause harm on account of their extensive use.

Heavy metals are dangerous because they tend to bio-accumulate. Bio-accumulation occurs when the toxic chemical is taken into the body faster than it can be excreted. Lead can have an adverse impact on mental development in infants and children. Lead may also be a factor in behavioural problems. Heavy metal poisoning could result, for instance, from drinking-water contamination.

Lead is the most prevalent heavy metal contaminant. As a component of tetraethyl lead, $(\text{CH}_3\text{CH}_2)_4\text{Pb}$, it was used extensively in 'leaded petrol' from the 1930s-1970s. Although the use of leaded petrol has been phased out, soils next to roads can have high lead concentrations – see **Figure 1**. Lead-based paints were another early source of lead pollution but their use is now banned in the UK. **Figure 2** opposite shows how the amount of lead used in paint and petrol in the USA changed over the 20th century.

Figure 1

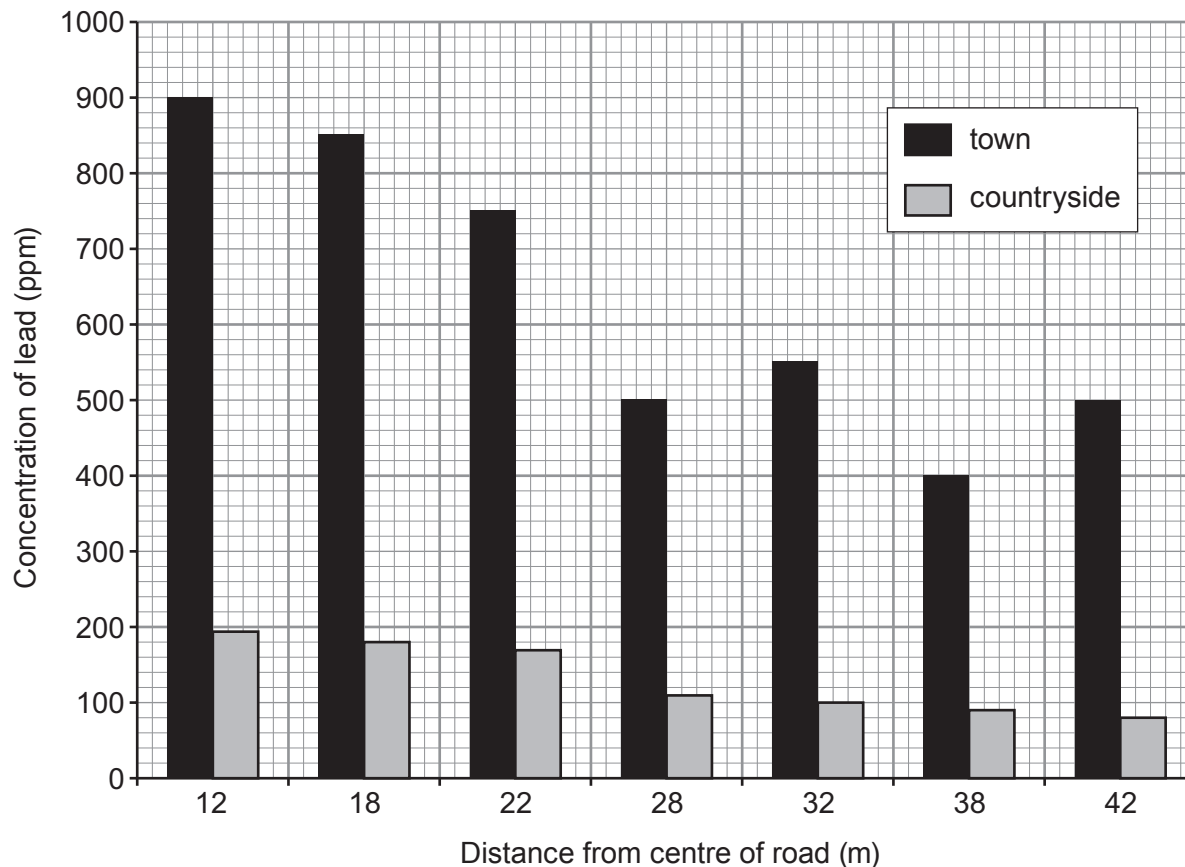
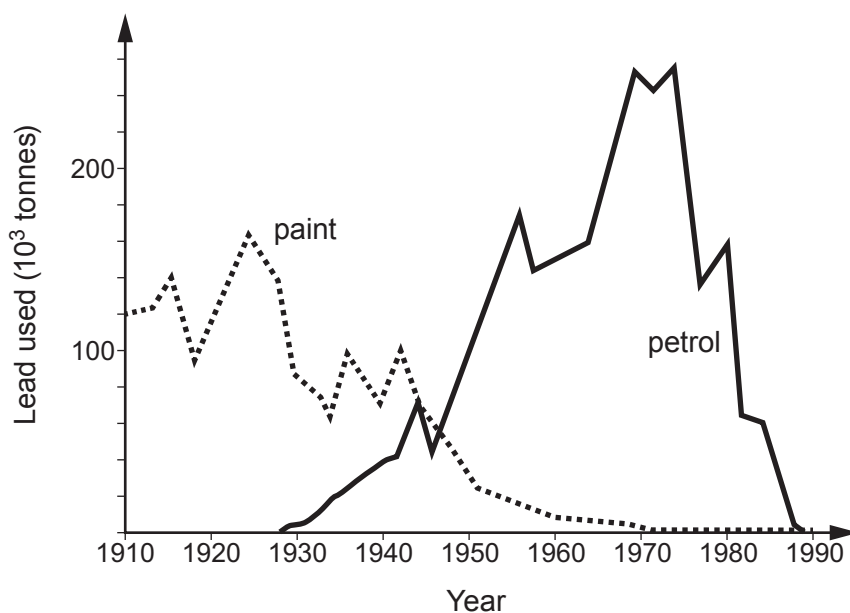


Figure 2



- (a) (i) Put a tick (✓) in the box next to the statement which **best** describes why a single definition of a 'heavy metal' is not accepted by all scientists. [1]

metallurgists, physicists and chemists do not trust each other

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metallurgists, physicists and chemists are not concerned with the fact that heavy metals are toxic

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metallurgists, physicists and chemists are concerned with different properties of heavy metals

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metallurgists, physicists and chemists study different heavy metals

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- (ii) Since the 1970s the use of lead water pipes has been prohibited across Europe. Explain why this is the case. [1]

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